

Minkyu Jeon

Ph.D. Candidate in Computer Science, Princeton University

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Research Profile

Ph.D. candidate in Computer Science at Princeton University, advised by Prof. Ellen D. Zhong. I build **scientific AI systems**: agentic, auditable workflows that coordinate models, scientific software, and high-performance computing for multi-step discovery. My current focus is agentic AI for autonomous research; de novo protein binder design is the application domain and testbed for closed-loop generation, verification, and compute allocation.

Research interests: scientific AI systems; agentic AI; autonomous research; inference-time scaling; tool-using scientific agents; adaptive compute allocation; AI for biology; de novo protein design; protein generative models; cryo-EM reconstruction; 3D generative modeling; representation learning.

Research Directions

Scientific AI agents	Closed-loop systems that combine generative models, scientific tools, validation routines, compute orchestration, structured evidence, and auditable decision traces.
Autonomous research	Inference-time scaling methods where the system improves outcomes by deciding what scientific action to run next, rather than only scaling a static model call.
Protein design	De novo binder design as a high-impact application for agentic discovery, integrating generators, redesign tools, structure predictors, interface diagnostics, and candidate-panel selection.
Cryo-EM reconstruction	Generative and representation learning methods for heterogeneous cryo-EM reconstruction and 3D molecular structure analysis.

Selected Research Projects

T-ReX: Agentic Inference-Time Scaling for De Novo Protein Design *2026–present*
Princeton University

- Developing a closed-loop scientific AI system that treats de novo binder design as target-adaptive compute allocation across registered generation, redesign, scoring, and validation routes.
- Designed an LLM Planner/Supervisor over structured **EvidenceSummary** records; deterministic software validates schemas, enforces bounds, schedules workers, and records evidence trails.
- Built evaluation workflows for structural success, Foldseek structural uniqueness, MMseqs2 sequence novelty, AlphaFold-style interface diagnostics, ESMFold2/Boltz-family post hoc scoring, affinity estimates, and candidate-panel ranking.

Heterogeneous Cryo-EM Reconstruction *2023–present*
Princeton University

- Developed generative, amortized, self-distilled, and transformer-based methods for recovering continuous conformational heterogeneity from noisy cryo-EM particle images.
- Led CryoBench, a NeurIPS 2024 Spotlight benchmark, and co-led CryoHype, a CVPR 2026 transformer-hypernetwork method; developed evaluation workflows for reconstruction fidelity, latent organization, coverage, and robustness.

Education

Princeton University *2023–present*
Ph.D. Candidate, Computer Science Princeton, NJ

Korea University *2020–2023*
M.S., Computer Science and Engineering Seoul, Korea

Konkuk University *2014–2020*
B.S., Applied Statistics and Computer Engineering, double major Seoul, Korea

Publications

* denotes equal contribution.

CVPR 2026 **CryoHype: Reconstructing a Thousand Cryo-EM Structures with Transformer-Based Hypernetworks**

Jeffrey Gu^{*}, Minkyu Jeon^{*}, Ambri Ma, Serena Yeung-Levy, Ellen D. Zhong
[paper](#)

arXiv 2026 **ConforNets: Latents-Based Conformational Control in OpenFold3**

Minji Lee, Colin Kalicki, Minkyu Jeon, Aymen Qabel, Alisia Fadini, Mohammed AlQuraishi
[paper](#)

NeurIPS MLSB 2025 **Separating Signal from Noise: A Self-Distillation Approach for Amortized Heterogeneous Cryo-EM Reconstruction**

Minkyu Jeon^{*}, Jeffrey Gu^{*}, Ambri Ma, Serena Yeung-Levy, Vincent Sitzmann, Ellen D. Zhong
Oral presentation. [paper](#)

NeurIPS 2024	CryoBench: Diverse and Challenging Datasets for the Heterogeneity Problem in Cryo-EM Minkyu Jeon , Rishwanth Raghu, Miro Astore, Geoffrey Woollard, Ryan Feathers, Alkin Kaz, Sonya M. Hanson, Pilar Cossio, Ellen D. Zhong Spotlight. paper / project
NeurIPS 2022	SageMix: Saliency-Guided Mixup for Point Clouds Sanghyeok Lee [*] , Minkyu Jeon [*] , Injae Kim, Yunyang Xiong, Hyunwoo J. Kim paper / code
ECCV 2022	<i>k</i>-SALSA: <i>k</i>-Anonymous Synthetic Averaging of Retinal Images via Local Style Alignment Minkyu Jeon , Hyeonjin Park, Hyunwoo J. Kim, Michael Morley, Hyunghoon Cho paper / code
JMIR 2024	A Multivariable Prediction Model for Mild Cognitive Impairment and Dementia: Algorithm Development and Validation Sarah Soyeon Oh, Bada Kang, Dahye Hong, Jennifer Ivy Kim, Hyewon Jeong, Jinyeop Song, Minkyu Jeon
Information Sciences 2023	Randomly Shuffled Convolution for Self-Supervised Representation Learning Youngjin Oh [*] , Minkyu Jeon [*] , Dohwan Ko, Hyunwoo J. Kim paper / code
IEEE Access 2021	Learning to Balance Local Losses via Meta-Learning Seungdong Yoa, Minkyu Jeon , Youngjin Oh, Hyunwoo J. Kim paper
In Submission	CHIMERA: Adaptive Cache Injection and Semantic Anchor Prompting for Zero-Shot Image Morphing with Morphing-Oriented Metrics Dahyeon Kye, Jaehun Sung, Minkyu Jeon , Jihyong Oh paper / project
arXiv 2025	R3eVision: A Survey on Robust Rendering, Restoration, and Enhancement for 3D Low-Level Vision Weeyoung Kwon, Jaehun Sung, Minkyu Jeon , Chanho Eom, Jihyong Oh paper

Research Experience

Prescient Design, Genentech Contractor	<i>Nov 2025–Mar 2026</i> New York, NY
• Worked on protein representation learning and generative modeling, with emphasis on modern 3D generative methods including diffusion and flow matching. • Investigated limitations of voxel-space and structure-space generative models and developed systematic workflows for diagnosing generation quality, failure modes, and target-dependent behavior. • Connected model-level analysis with practical protein-design evaluation, including structural quality, interface plausibility, robustness, and downstream candidate selection.	
Prescient Design, Genentech Research Intern	<i>Jun 2025–Aug 2025</i> New York, NY
• Led a research project on understanding and improving generative models in complex 3D spaces, with applications to protein generation and design. • Designed controlled experiments and evaluation procedures for relating generative-model behavior to practical scientific outcomes across diverse target settings.	
AI4ALL Instructor	<i>Jul 2024–Aug 2024</i> Princeton, NJ
• Taught artificial intelligence to high school students and mentored team projects on deep learning for medical image analysis.	
Broad Institute of MIT and Harvard Associate Computational Biologist; advisor: Dr. Hyunghoon Cho	<i>Jan 2023–Aug 2023</i> Cambridge, MA
• Led a project on privacy-preserving retinal image synthesis using diffusion-based generative models. • Developed generative modeling and evaluation workflows connecting image fidelity, privacy preservation, and downstream scientific utility.	
Broad Institute of MIT and Harvard Visiting Graduate Student; advisor: Dr. Hyunghoon Cho	<i>Sep 2021–Apr 2022</i> Cambridge, MA

- Led GAN- and GAN-inversion-based research on privacy-preserving retinal image generation, resulting in an ECCV 2022 publication.

Korea University
 Research Intern; advisor: Prof. Hyunwoo J. Kim

Jan 2020–Aug 2020
 Seoul, Korea

- Conducted meta-learning research for adaptive layer-wise optimization and co-authored an IEEE Access publication.
- Designed algorithms, implemented experiments, and contributed to the related work, methods, and experiments sections.

MakinaRocks
 ML Research Engineer Intern

Mar 2019–Aug 2019
 Seoul, Korea

- Conducted anomaly-detection research using variational autoencoders and latent-space regularization.

Korea Institute of Science and Technology
 ML Research Intern; advisor: Dr. Yong Moo Kwon

Jul 2018–Dec 2018
 Seoul, Korea

- Developed components of an active chatbot system using rule-based natural language processing and deep learning for situation-aware interaction.

Awards and Service

2024–present	Asan Foundation Biomedical Science Scholarship. Three-year doctoral scholarship.
2023–present	Reviewer. CVPR (2026), ECCV (2026), ICLR (2025, 2026), NeurIPS (2025, 2026), NeurIPS Workshops (2023, 2024), and Information Sciences (2024).
Aug 2024	Invited talk. Flatiron Institute, Simons Foundation.
Feb 2024	Invited talk. CMU MetaMobility Lab.
Sep 2019	Dean’s List. Department of Applied Statistics, Konkuk University.
Sep 2019	Konkuk University Scholarship. 40% tuition scholarship.
Jul 2016	Certificate of Commendation. Republic of Korea Marine Corps.

Teaching

Fall 2024	Teaching Assistant, COS 302: Mathematics for Numerical Computing and Machine Learning, Princeton University.
Spring 2021	Teaching Assistant, COSE 361: Artificial Intelligence, Korea University.
Fall 2020	Teaching Assistant, AAA 712: AI Security, Korea University.
Fall 2020	Tutorial Teaching Fellow, Deep Learning, Korea University.

Technical Skills

Scientific AI systems	Agentic scientific workflows, autonomous research loops, evidence-based planning, inference-time scaling, compute allocation, deterministic execution layers, reproducible experiment traces, Linux, Slurm, multi-GPU/HPC workflows, experiment orchestration.
AI for biology / structure	De novo protein design, protein structure evaluation, interface-confidence diagnostics, candidate-panel prioritization, AlphaFold-style confidence diagnostics, Foldseek, MMseqs2, structural clustering, sequence/structure novelty analysis, cryo-EM reconstruction, molecular structure analysis.
Generative ML	Diffusion models, flow matching, generative adversarial networks, variational autoencoders, hypernetworks, representation learning, self-distillation, 3D vision, inverse problems, scientific benchmarking.
Programming	Python, PyTorch, TensorFlow, Keras, R, C, Git.

Additional Experience

I-Corps Program Received \$35,000 in government support for customer discovery and technology-based startup exploration.	<i>May 2021–Aug 2021</i>
Republic of Korea Marine Corps Military service, Yeonpyeong Island, South Korea; received a commendation for communication and performance during early-warning situations.	<i>Mar 2015–Dec 2016</i>